

Decoding Customer Behaviour

A Data Cleaning & Business Insights Report Using Python and SQL

BUSINESS PROBLEM	OBJECTIVE
- Raw retail transaction data is rarely analysis-ready — it contains missing values, inconsistent formatting, and unstructured text fields - Missing review ratings make product-quality analysis unreliable if left unaddressed - Inconsistent column naming and mixed-case headers create errors in downstream Python and SQL code - Text-based fields (e.g. purchase frequency) cannot be used directly in numeric or trend calculations - Without clean, structured data, decision-makers cannot confidently answer basic questions about who customers are, how they behave, and where revenue actually comes from - This leads to guesswork in marketing spend, inventory planning, and retention strategy	- Clean and prepare a raw retail customer-transactions dataset using Python (pandas) - Handle missing values intelligently using category-level medians rather than blunt averages - Standardize column names and structure for consistent, error-free downstream use - Engineer new features — <i>age_group</i> and <i>purchase_frequency_days</i> — to unlock deeper analysis - Remove redundant columns to produce a lean, analysis-ready dataset - Apply structured SQL analysis to answer ten real business questions on revenue, discounts, subscriptions, shipping, and loyalty - Deliver evidence-based insights that directly inform retail strategy

APPROACH: PART A — DATA CLEANING (PYTHON)

All cleaning was performed using the pandas library in a Google Colab environment, transforming the raw CSV into a reliable, analysis-ready dataset before any business questions were addressed.

STEP	WHAT WAS DONE
Load & Inspect	Loaded the raw CSV into a pandas DataFrame; reviewed shape, data types, and completeness using <i>df.info()</i> and <i>df.isnull().sum()</i>
Handle Missing Values	Filled missing <i>Review Rating</i> values using the median rating of each product category (via <i>groupby</i> + <i>transform</i>), avoiding distortion from outliers
Standardize Columns	Converted all column names to lowercase, replaced spaces with underscores, and renamed <i>purchase_amount_usd</i> to <i>purchase_amount</i>
Age Group Feature	Created an <i>age_group</i> column using quartile binning (<i>pd.qcut</i>) into Young Adult, Adult, Middle-aged, and Senior bands
Purchase Frequency Feature	Mapped text values like "Weekly" or "Monthly" to a numeric <i>purchase_frequency_days</i> column for trend analysis
Remove Redundancy	Confirmed <i>promo_code_used</i> always matched <i>discount_applied</i> and dropped it, producing a leaner dataset

APPROACH: PART B — SQL BUSINESS ANALYSIS

With the dataset cleaned, ten business questions were answered by running SQL queries against the *customer* table — using GROUP BY aggregation, subqueries, CASE logic, Common Table Expressions (CTEs), and window functions such as *ROW_NUMBER()* — to turn raw records into decision-ready insights.

KEY BUSINESS QUESTIONS & INSIGHTS

BUSINESS QUESTION		KEY INSIGHT DELIVERED
Q1	Revenue by gender	Reveals which gender segment drives more revenue
Q2	Discount users spending above average	Flags a high-value, price-aware customer segment
Q3	Top 5 rated products	Identifies products worth featuring or promoting
Q4	Standard vs Express shipping spend	Tests link between shipping choice and order value
Q5	Subscriber vs non-subscriber spend	Quantifies the value of subscription programs
Q6	Top 5 most discounted products	Flags products overly reliant on discounting
Q7	New / Returning / Loyal segmentation	Builds a customer loyalty funnel
Q8	Top 3 products per category	Supports category-level inventory & promotion
Q9	Repeat buyers vs subscription	Tests if loyalty correlates with subscribing
Q10	Revenue by age group	Shows which age band drives the most revenue

TOOLS USED	DATASET	DELIVERABLE
Python (pandas), SQL, Google Colab	Retail Customer Shopping Transactions	10 Business-Ready SQL Insights

CONCLUSION

Decoding customer behaviour began with disciplined data cleaning: filling missing review ratings using category-level medians, standardizing column names, and engineering two new features — *age_group* and *purchase_frequency_days* — that unlocked deeper analysis. On this clean foundation, ten SQL-driven business questions were answered, spanning revenue by gender and age, the impact of discounts and subscriptions, shipping preferences, top-rated and top-selling products, and customer loyalty segmentation. Together, these findings give a rounded, evidence-based view of who the customers are, how they behave, and where the business should focus its attention.